

COSC/MATH 4340: Numerical Methods for Differential Equations

Numerical Linear Algebra (NLA)

Class: COSC/MATH 4340

Semester: Spring 2024

Instructor: Dane Taylor

Office: Eng 4087

Email: dane.taylor@uwyo.edu





Overview

Consider $A\mathbf{x} = \mathbf{b}$, or

$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ A_{21} & \ddots & A_{2n} \\ \vdots & & \vdots \\ A_{n1} & & A_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

This has a solution $\mathbf{x} = A^{-1}\mathbf{b}$, but it is very expensive (perhaps impossible) to compute A^{-1} , which should be avoided.

Here we study different ways to find $\mathbf{x}$



Gaussian Elimination

Definition

Gaussian Elimination finds a solution using 2 steps

- ▶ Transform $A\mathbf{x} = \mathbf{b}$ into an equivalent system $U\mathbf{x} = \mathbf{g}$ that has the same solution $\mathbf{x}$, but which can be easily and efficiently solved.
- ▶ Solve $U\mathbf{x} = \mathbf{g}$ using back substitution: first solve x_n , then $x_{n-1}, x_{n-2}, \dots$

(See notes and code for example.)



Gaussian Elimination

Step 1: Implement a sequence of row operations that transforms A into U one row at a time.

- ▶ Let $A^{(1)}\mathbf{x} = \mathbf{b}^{(1)}$ denote the original equation.
- ▶ Define the row multipliers $m_{ik} = A_{ik}^{(k)} / A_{kk}^{(k)}$ for $i = k + 1, \dots, n$
- ▶ Then define a sequence of equivalent linear systems $A^{(k)}\mathbf{x} = \mathbf{b}^{(k)}$ with

$$A_{ij}^{(k+1)} = A_{ij}^{(k)} - m_{ik}A_{kj}^{(k)}, \quad i, j \in \{k + 1, \dots, n\}$$

$$b_i^{(k+1)} = b_i^{(k)} - m_{ik}b_k^{(k)}, \quad i, j \in \{k + 1, \dots, n\}$$

- ▶ After transforming all n rows, define $U = A^{(n)}$ and $\mathbf{g} = \mathbf{b}^{(n)}$.

Gaussian Elimination

Step 1 yields an equivalent system $U\mathbf{x} = \mathbf{g}$ that uses an upper triangular matrix $U = A^{(n)}$

$$\begin{bmatrix} A_{11}^{(1)} & \dots & & & A_{1n}^{(1)} \\ 0 & A_{22}^{(2)} & & & A_{2n}^{(2)} \\ \vdots & & \ddots & & \vdots \\ 0 & \dots & & A_{n-1,n-1}^{(n-1)} & A_{n-1,n}^{(n)} \\ 0 & \dots & & & A_{nn}^{(n)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(2)} \\ \vdots \\ b_{n-1}^{(n-1)} \\ b_n^{(n)} \end{bmatrix}$$

Here, the x_j are the same, but the $a_{ij}^{(i)}$ and $b_i^{(i)}$ have changed.



Gaussian Elimination

Step 2: Solve $U\mathbf{x} = \mathbf{g}$ for the x_j using back substitution

$$x_n = g_n / U_{nn}$$

$$x_{n-1} = [g_{n-1} - U_{n-1,n}x_n] / U_{n-1,n-1}$$

$$x_{n-2} = \dots$$

$$x_k = [g_k - \sum_{j=k+1}^n U_{k,j}x_j] / U_{kk}$$

Note that you solve the x_k in reverse order: $n, n-1, n-2, \dots$



LU Decomposition

Definition

The **LU Decompositions** of a matrix A is given by $A = LU$, where $U = A^{(n)}$ is the **upper triangular** matrix resulting from Gaussian elimination and L is a **lower triangular** matrix with 1's on the diagonal and the lower-triangular entries encode the multipliers:

$$L = \begin{bmatrix} 1 & 0 & \dots & & 0 \\ m_{21} & 1 & 0 & & \vdots \\ m_{31} & m_{32} & \ddots & \ddots & \\ \vdots & \dots & \ddots & 1 & 0 \\ m_{n1} & \dots & & m_{n,n-1} & 1 \end{bmatrix}$$



LU Decomposition

Definition

The **computational complexity** of an algorithm is the maximum of the following two counts

- ▶ the total number of additions and subtractions
- ▶ the total number of multiplications and divisions

Theorem

The computational complexity for constructing an LU decomposition using Gaussian elimination is $\mathcal{O}(n^3)$

(See notes for proof.)



Error Analysis

We now study the impact of rounding errors on solutions to linear systems, starting with some background info:

Definition

A **vector norm** $||\mathbf{x}||$ is a scalar function such that

$$||\mathbf{x}|| \geq 0 \quad (1)$$

$$||\alpha\mathbf{x}|| = |\alpha| ||\mathbf{x}|| \quad (2)$$

$$||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}|| \quad (3)$$

$$||\mathbf{x} - \mathbf{z}|| \leq ||\mathbf{x} - \mathbf{y}|| + ||\mathbf{y} - \mathbf{z}|| \quad (4)$$

$$|||\mathbf{x}|| - ||\mathbf{y}||| \leq ||\mathbf{x} - \mathbf{y}|| \quad (5)$$



Error Analysis

Definition

A **matrix norm** $\|A\|$ is a scalar function such that

$$\|A\| \geq 0 \quad (6)$$

$$\|\alpha A\| = |\alpha| \|A\| \quad (7)$$

$$\|A + B\| \leq \|A\| + \|B\| \quad (8)$$

$$\|AB\| \leq \|A\| \|B\| \quad (9)$$

$$\|A\mathbf{x}\|_{\nu} \leq \|A\| \|\mathbf{x}\|_{\nu} \quad (10)$$

where $\|\mathbf{x}\|_{\nu}$ is some vector norm.

(See notes for compatible matrix and vector norms.)



Error Analysis

The condition number measures the sensitivity of matrix-related problems to perturbations.

Definition

Let $|| \cdot ||$ denote a matrix norm, then the **condition number** $\text{cond}(A)$ of a matrix A with respect to that norm is given by:

$$\text{cond}(A) = ||A|| ||A^{-1}||.$$

Theorem

$$\text{cond}(A) \geq 1.$$

(See notes for proof.)



Error Analysis

Consider how a perturbation $\delta \mathbf{b}$ to $\mathbf{b}$ will cause a perturbation $\delta \mathbf{x}$ on the solution $\mathbf{x}$ to $A\mathbf{x} = \mathbf{b}$.

Theorem

Let $A\mathbf{x} = \mathbf{b}$ and $A(\mathbf{x} + \delta \mathbf{x}) = \mathbf{b} + \delta \mathbf{b}$. Then $A\delta \mathbf{x} = \delta \mathbf{b}$, $\delta \mathbf{x} = A^{-1}\delta \mathbf{b}$ and we have the bound

$$\frac{1}{\text{cond}(A)} \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \leq \frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \text{cond}(A) \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|}.$$

(See notes for proof.)

Error Analysis

Theorem

Consider $A\mathbf{x} = \mathbf{b}$ with nonsingular A with perturbations δA to A and $\delta \mathbf{b}$ to $\mathbf{b}$ and assume $\|\delta A\| < 1/\|A^{-1}\|$. Then $A + \delta A$ is also nonsingular and the solution to

$$(A + \delta A)(\mathbf{x} + \delta \mathbf{x}) = (\mathbf{b} + \delta \mathbf{b})$$

satisfies

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond}(A)}{1 - \text{cond}(A) \frac{\|\delta A\|}{\|A\|}} \left[\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|} \right]$$



Error Analysis for Gaussian Elimination

Theorem

Define $\rho = \frac{1}{\|A\|_\infty} \max_{i,j,k} |A_{ij}^{(k)}|$. Then

- ▶ GE gives $LU = A + E$ and the rounding error is bounded by $\|E\|_\infty \leq n^2 \rho \|A\|_\infty \epsilon_{mach}$
- ▶ The approximated solution $\tilde{\mathbf{x}}$ using GE satisfies $(A + \delta A)\tilde{\mathbf{x}} = \mathbf{b}$ where:

$$\frac{\|\delta A\|}{\|A\|} \leq 1.01(n^3 - 3n^2)\rho\epsilon_{mach}$$

$$\frac{\|\mathbf{x} - \tilde{\mathbf{x}}\|}{\|\mathbf{x}\|} \leq \frac{\text{cond}(A)}{1 - \text{cond}(A) \frac{\|\delta A\|}{\|A\|}} \frac{\|\delta A\|}{\|A\|}.$$